3. $$\text{Error} = \frac{(x-x_0)(x-x_1)(x-x_2)}{6} e^{c_x}$$

$x_1 = x_0 + h$

$x_2 = x_0 + 2h$

where $x \in [x_0, x_2]$ and x $[0,1]$

Now,

$$|\text{Error}| \leq \left|(x-x_0)(x-x_1)(x-x_2)\right| \frac{e}{6}$$

$$\leq \left|(x-x_0)(x-x_0-h)(x-x_0-2h)\right| \frac{e}{6}$$

Let $y = x - x_0 - h$

$$\leq \left|(y+h)(y)(y-h)\right| \frac{e}{6}$$

$$\leq \left|y^3 - h^2 y\right| \frac{e}{6} \quad -(1)$$

So, from (1), we get

$$g(y) = y^3 - h^2 y$$

$$g'(y) = 3y^2 - h^2 = 0$$

$$\Rightarrow y = \pm \frac{h}{\sqrt{3}}$$

$$\Rightarrow g\left(\frac{h}{\sqrt{3}}\right) = \left|\frac{h}{\sqrt{3}}\left(\frac{h^2}{3} - h^2\right)\right|$$

$$= \frac{2h^3}{3\sqrt{3}}$$

$$\Rightarrow \boxed{|\text{Error}| \leq \frac{2}{3\sqrt{3}} h^3 \frac{e}{6}}$$